BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2025

CHEMISTRY

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Objective Type) - Group - I

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. In an irreversible reaction, dynamic equilibrium:

(A) Never established
(B) Establishes readily
(C) Establishes before the completion of reaction
(D) Establishes after the completion of reaction

Q2. Which one of the following is a Lewis base :

(A) NH_3
(B) BF_3
(C) H^+
(D) $AlCl_3$

Q3. $K_4[Fe(CN)_6]$ is a:

(A) Acidic salts
(B) Double salts
(C) Complex salts
(D) Mixed salts

Q4. Which salt is used for manufacture of explosives, plastics and other chemicals:

(A) N_2ClO_3
(B) N_2SiO_3
(C) Na_2SO_4
(D) Na_2CO_3

Q5. Anthracite contains the carbon content:

(A) 80%
(B) 70%
(C) 90%
(D) 60%

Q6. Substitution reaction is the characteristics of:

(A) Alkanes
(B) Alkenes
(C) Alkynes
(D) CO_2

Q7. Which one of the following vitamins is water soluble :

(A) Vitamin A
(B) Vitamin D
(C) Vitamin E
(D) Vitamin C

Q8. A good example of reducing sugar is:

(A) Glucose
(B) Maltose
(C) Sucrose
(D) Starch

Q9. Which one of these pollutants is not found in car exhaust fumes :

(A) CO
(B) O_3
(C) NO_2
(D) SO_2

Q10. Which one of the following gases is used to destroy harmful bacteria in water:

(A) Iodine
(B) Chlorine
(C) Fluorine
(D) Bromine

Q11. The type of hepatitis which can be caused by contaminated water is :

(A) E
(B) C
(C) D
(D) B

Q12. Which one of the following is not fraction of residual oil:

(A) Paraffin wax
(B) Asphalt
(C) Fuel oil
(D) Petroleum coke

Chemistry (Subjective) (PART - I) - Group - I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Write two macroscopic characteristics of forward reaction.

(ii) Write the equilibrium constant expression for the reaction : $2SO_2 + O_2 \rightleftharpoons 2SO_3$

(iii) Define chemical equilibrium state.

(iv) Write any two physical properties of acid.

(v) Write reaction of acid with sulphides (Give equation).

(vi) Write formula of sodium hydroxide and potassium hydroxide.

(vii) How are alkyl halides reduced?

(viii) Define hydrocarbons. Give an example.

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Why and how carbon completes its octet?

(ii) Define destructive distillation.

(iii) What is an ester group? Write down the formula of ethyl acetate.

(iv) Write down the general formula of lipids.

(v) Write the names of two fatty acids with their formulae.

(vi) Define pollutants and contaminants.

(vii) How is the temperature of atmosphere maintained?

(viii) Write the health problems created by SO_2.

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Water dissolves sugar and alcohol, justify it.

(ii) Define soft water and hard water.

(iii) What is meant by leaching process?

(iv) Why are the non-polar compounds insoluble in water?

(v) How $NaHCO_3$ is converted to Na_2CO_3?

(vi) Describe froth flotation process.

(vii) How is roasting carried out in extraction of metal?

(viii) How is ammonia prepared for the manufacture of urea?

Chemistry (Subjective) (PART - II) - Group - I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

Q5(a). (Marks 5) Define salts. Explain preparation of soluble salts.

Q5(b). (Marks 4) Differentiate between homocyclic and alicyclic compounds.

Q6(a). (Marks 5) Give five uses of ethylene.

Q6(b). (Marks 4) Explain the sources and uses of protein.

Q7(a). (Marks 5) Define refining of petroleum and explain fractional distillation of petroleum.

Q7(b). (Marks 4) Give four effects of acid rain.

GROUP – II

Chemistry (Objective Type) - Group - II

Time Allowed: 15 Min. | **Maximum Marks:** 12

> Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Which one of the following is a reducing sugar:

(A) Maltose
(B) Glucose
(C) Sucrose
(D) Starch

Q2. Dysentery is a disease of:

(A) Heart
(B) Liver
(C) Stomach
(D) Intestine

Q3. The formula of bleaching powder is:

(A) $Ca(OH)_2$
(B) Ca(OCl)
(C) $CaCl_2$
(D) Ca(OCl)Cl

Q4. Just above the earth's surface is:

(A) Mesosphere
(B) Stratosphere
(C) Thermosphere
(D) Troposphere

Q5. Concentration is a:

(A) Mixing technique
(B) Separating technique
(C) Boiling technique
(D) Cooling technique

Q6. At dynamic equilibrium:

(A) The reaction stops to proceed
(B) The amounts of reactants and products are equal
(C) The speeds of forward and reverse reactions are equal
(D) The reaction can no longer be reversed

Q7. The order of reactivity of hydrogen halides with alkenes is:

(A) HI > HBr
(B) HBr > HI
(C) HCl > HBr
(D) HBr < HCl

Q8. Which is the king of chemicals :

(A) HNO_3
(B) HCl
(C) CH_3COOH
(D) H_2SO_4

Q9. Who proposed the name of vitamins:

(A) Watson
(B) Lewis
(C) F. Crick
(D) Funk

Q10. Which one of the following is not a fossil fuel:

(A) Coal
(B) Natural gas
(C) Biogas
(D) Petroleum

Q11. Which one of the following is a Lewis base :

(A) NH_3
(B) BF_3
(C) H^+
(D) $AlCl_3$

Q12. Which of the following gas is used to destroy harmful bacteria in water:

(A) Iodine
(B) Fluorine
(C) Chlorine
(D) Bromine

Chemistry (Subjective) (PART - I) - Group - II

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Write two macroscopic characteristics of dynamic equilibrium.

(ii) Differentiate between forward and reverse reactions.

(iii) Why reversible reactions never complete?

(iv) Define salts with an example.

(v) Write name of two acids used in the manufacturing of fertilizers.

(vi) Define Lewis concept of acids and bases.

(vii) Why are the alkanes called "paraffins"?

(viii) Define closed chain or cyclic hydrocarbons.

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

(i) What are alkyl radicals? Give their general formula.

(ii) What is the importance of natural gas?

(iii) Define molecular formula. Give an example.

(iv) Write down two uses of lipids.

(v) What are amino acids? Give general formula.

(vi) Define primary pollutants and give an example.

(vii) Why does acid rain damage buildings?

(viii) Why does temperature decrease in troposphere?

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Why is the water molecule polar?

(ii) How does sodium zeolite soften the water?

(iii) What do you mean by fluorosis?

(iv) What do you mean by boiler scales? How are they removed?

(v) What do you mean by anode mud?

(vi) How is CO_2 prepared in the Solvay's process?

(vii) What happens when ammonium carbamate is heated with steam?

(viii) What role is played by pine oil in the froth flotation process?

Chemistry (Subjective) (PART - II) - Group - II

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

Q5(a). (Marks 5) Define an acid and a base according to Bronsted Lowry concept and justify with example.

Q5(b). (Marks 4) Explain four general characteristics of organic compounds.

Q6(a). (Marks 5) Explain five uses of acetylene.

Q6(b). (Marks 4) Explain oligosaccharides.

Q7(a). (Marks 5) How crude oil is refined? Explain two important fractions of petroleum along their usage.

Q7(b). (Marks 4) Write the four effects of ozone depletion.